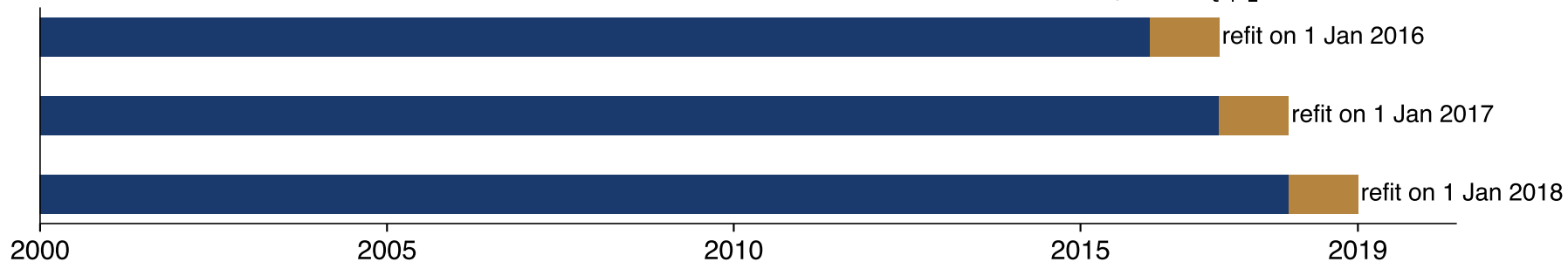
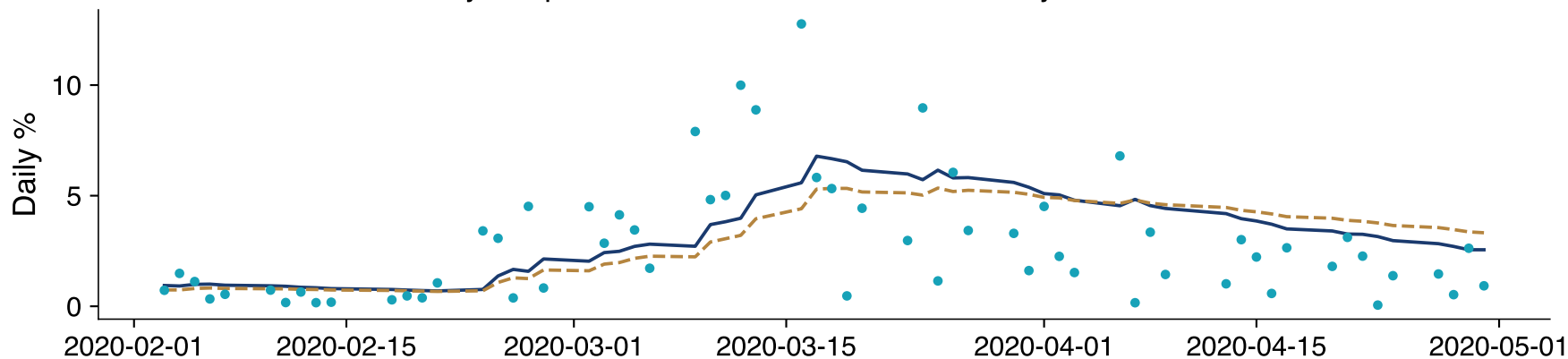


Forecast made at the close of day  $t$  with data up to  $t$ ; evaluated against  $r_{t+1}^2$



February to April 2020: the forecast reacts one day after each shock



- Estimation sample (expanding window, from 3 Jan 2000)
- Forecast year: parameters fixed, variance updated daily
- $\sqrt{h_t}$ , GARCH-t (daily %)
- $\sqrt{h_t}$ , EWMA
- $|r_t|$  (the square root of the proxy  $r_t^2$ )